

Ian Salandy

Systems Pressure Architect · Performance Engineering · Distributed Systems

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SUMMARY

Systems Pressure Architect with 17+ years defining technical direction for large-scale distributed infrastructure at global enterprises. Expert in capacity planning, cost-to-serve modeling, and system-level performance engineering across multi-billion event workloads. Trusted advisor to engineering leadership on system risk, scalability strategy, and infrastructure investment decisions.

Performance pressure is not latency — it is the accumulated stress from traffic shape, resource saturation, coupling density, operational fragility, and economic misalignment, manifesting before any single metric crosses a threshold. My work maps where that pressure builds and traces its propagation paths before it surfaces as outages or cost overruns.

Methodology formalised as the **PPI-F™ (Performance Pressure Index) Framework** — structured governance across Performance, Production Readiness, Infrastructure Efficiency, and Failure Resilience. Advisory through **KPI99**.

CORE COMPETENCIES

- Server & Compute Infrastructure Technical Direction
 - Hardware Lifecycle Planning & Capacity Strategy
 - Infrastructure Cost Optimization & Efficiency Modeling
 - Scalability, Reliability & Performance Engineering at Hyperscale
 - Compute, Storage & Network Systems Architecture
 - Technical Specification Definition & Standards Governance
 - Cross-Functional Leadership & Executive Influence
 - Operational Excellence, Fleet Management & Incident Response
 - Deployability & Serviceability Strategy
 - Platform Engineering Alignment & Roadmap Collaboration
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PPI-F™ FRAMEWORK ALIGNMENT

The four pillars of the PPI-F™ Framework map directly to the operating responsibilities of Google's ServerOps Principal Engineer:

PPI-F™ Pillar	ServerOps Application
Performance	Define server hardware specs, performance metrics & throughput targets
Production Readiness	Deployability, serviceability & NPI compatibility across data center designs
Infrastructure Efficiency	Balance cost, efficiency & technology innovation across the global fleet
Failure Resilience	Capacity planning, incident response & fleet maintenance reliability

PROFESSIONAL EXPERIENCE

Salesforce — Acting Principal Architect / Staff Performance Engineer

2021 – Present

- Set overall technical direction for distributed server infrastructure processing billions of events annually across multi-region compute, storage, and network deployments.
- Defined system-level scalability specifications — performance metrics, throughput targets, and reliability standards — across compute, storage, and query infrastructure, improving fleet throughput by 35% under production-equivalent load.
- Built cost-to-serve models that informed infrastructure consolidation, reducing annual spend by ~\$750K without degrading SLAs.
- Developed predictive workload models forecasting infrastructure behavior under 2x–10x growth scenarios, enabling proactive fleet capacity investment decisions.
- Established cross-system diagnostics frameworks for bottleneck identification across heterogeneous data stacks (Spark, Trino, Kafka), directly improving reliability and reducing incident response time.
- Designed performance automation tooling enabling repeatable large-scale validation across deployment and upgrade cycles.
- Led benchmarking and regression governance, setting org-wide quality gates ensuring infrastructure stability across quarterly releases.
- Partnered with platform engineering and product operations leaders to align infrastructure specifications with hardware roadmap requirements and operational constraints.
- Assessed and selected technologies, frameworks, and tooling for optimal infrastructure operations and system validation.
- Presented infrastructure risk assessments, scaling constraints, and architectural investment recommendations to VP and C-suite stakeholders.

IBM — Principal Performance & Capacity Engineer

2016 – 2021

- Defined capacity planning and infrastructure scalability strategies for enterprise distributed systems supporting millions of daily transactions.
- Designed diagnostics frameworks to systematically analyze system behavior under sustained and peak load conditions.
- Built infrastructure efficiency models aligning compute and storage allocation to actual workload demand, reducing waste by 20%+.
- Led JVM and backend optimization initiatives reducing p99 latency and improving system reliability under variable load.
- Operated as internal performance architect across multiple product lines, standardizing performance criteria and escalation paths.

Accenture — Lead Performance Engineer

2009 – 2016

- Led performance architecture initiatives for large-scale enterprise systems across financial services, retail, and public sector clients.
- Designed load simulation and capacity stress frameworks to validate infrastructure scalability ahead of peak seasonal events.
- Diagnosed and resolved distributed system bottlenecks impacting production reliability and SLA attainment.
- Delivered performance readiness assessments and executive briefings for go-live decisions on multi-million-dollar programs.
- Mentored engineers and technicians, fostering a culture of continuous learning and technical excellence in infrastructure operations.

SELECTED IMPACT

Achievement	Outcome
Fleet-scale infrastructure architecture for billions of events	Multi-region reliability & sustained growth
35% throughput improvement via compute & storage optimization	Measurable fleet performance gain
\$750K+ annual infrastructure cost reduction	Efficiency modeling applied at fleet scale
Predictive capacity models for 2x–10x growth scenarios	Proactive fleet planning, failure prevention
Benchmarking & regression governance across quarterly releases	Infrastructure stability at org-wide scale

OPEN SOURCE & INDEPENDENT WORK

Scalebreaker — Gamified distributed systems architecture platform. Users design systems in a real-time flow interface and test against realistic performance metrics (throughput, latency, bottlenecks). Built to make architectural trade-offs legible through experience, not theory.

SudokuCV — This site. An interactive CV where Sudoku puzzles at increasing difficulty unlock progressively detailed sections of professional experience — a demonstration that how you present work is itself a signal about how you think.

TECHNOLOGIES

Infrastructure & Compute: Kubernetes (EKS), AWS (EC2, EMR, S3, CloudWatch), distributed compute clusters, server fleet capacity planning, telemetry & observability pipelines

Data & Messaging Systems: Apache Spark, Trino/Presto, Apache Kafka

Languages: Java, Python, SQL, Bash

Methodology: PPI-F™ Framework — Performance, Production Readiness, Infrastructure Efficiency, Failure Resilience. Advisory through KPI99.

EDUCATION

M.S. Computer Science — North Carolina State University | 2004 – 2006

Graduate research in distributed computing, parallel architectures, and performance modeling.

B.S. Computer Science — Morehouse College | 2000 – 2004

Foundation in algorithms, operating systems, networks, and software engineering principles.